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**Article**

**Keywords:**

**Posted Date:** July 5th, 2023

**DOI:** https://doi.org/10.21203/rs.3.rs-3109229/v1

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**Additional Declarations:** No competing interests reported.

**Version of Record:** A version of this preprint was published at Scientific Reports on November 11th, 2023.
See the published version at https://doi.org/10.1038/s41598-023-46787-5.

# Multiband Terahertz Metamaterial Perfect Absorber for Microorganisms Detection

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## ABSTRACT

We report a multi-resonant THz metamaterial perfect absorber-based biosensor in the working frequency range of 0.2 to 4THz for the sensing of microorganisms such as fungi, yeast, and pesticides. Nearly $100\%$ absorption is realized at $f_1 = 1.7THz, f_2 = 2.8THz, f_3 = 3.2THz, f_4 = 3.5THz$. To obtain optimum performance of the MPA, a mapping of amplitudes and shifts in the absorption resonance peaks with different structural parameters of the resonators is carried out. Investigations of the effect of the incidence angle of the THz wave on the absorption spectra of the MPA is also carried out. It is found that incidence angle does not impact the stability of the lower resonance absorption peak $(1.79THz)$. To better understand the mechanism of MPA, an equivalent circuit model using Advance Design System (ADS) software is developed. The calculated results through the model show similar trends as obtained in the simulations using CST. A significant frequency shift in the resonances is obtained with the application of microorganisms such as Penicillium chrysogenum (fungi), yeast, and pesticides (Imidacloprid, N, N-Diethyldithiocarbamate sodium salt trihydrate, Daminozide, N, N-Diethyldithiocarbamate sodium salt hydrate, and Dicofol) as testing analytes. The THz MPA-based biosensor has great potential to detect a very small change in the effective dielectric constant of the MPA environment therefore it can offer huge opportunities in label-free biosensing for future biomedical applications.

## Introduction

Sensors are vital part of modern technology due to their applications in imaging, medicine, food quality control, agriculture, defence, etc. Food safety and security, two foremost public health concerns[1–3], require fast and non-destructive inspection techniques. Due to fingerprints of pesticide and antibiotics in terahertz (THz) spectral range[4,5], THz spectroscopy facilitates non-contact, label-free, non-destructive inspection for food safety and control[6–11]. Moreover, the detection of microorganisms such as bacteria, viruses, and fungi is also very important for food security[12]. However, due to the typical size of $\lambda/100$, microorganisms are transparent at THz frequency leading to low scattering cross-section and are difficult to detect by the THz spectroscopy[13]. To overcome this constraint, metamaterial-based THz sensors have been proposed for the detection of microorganisms like bacteria, yeast, and molds[14], where metamaterial resonators are designed in such a way that the gap area of resonators becomes compatible with the size of microorganisms. Here, the resonance frequency of metamaterials is extremely sensitive to any changes in the dielectric constant of the gap area of the resonator[15,16]. Presently, extensive research is being carried out to develop metamaterial-based resonant devices such as modulators, sensors, polarization conversion devices, absorbers, etc.[17–19]. Among these devices, metamaterial perfect absorbers (MPAs) have fascinated worldwide attention due to their excellent performance and their potential applications[20,21].

MPAs are able to minimize the reflectivity and transmissivity close to zero. To further expand the application prospects of MPAs, the number of absorption frequency peaks and workable frequency bands need to be broadened. Several reports show that flexibility to work in a wide range of frequency band can be obtained by optimizing the dimensions of MPAs[22–28]. Zamzam *et. al*[29] designed an MPA with a metal-dielectric-metal multi-layer column with different heights of the dielectric layer, where they achieved two absorption peaks. Wang *et. al*[30] demonstrate a triple-band terahertz metamaterial absorber at 0.337, 0.496, and 0.718THz where they used three concentric square ring resonators. A metamaterial absorber consisting of three closed circular ring resonators on a polyethylene terephthalate substrate is investigated by Abdulkarim *et. al*[31]. The absorption mechanism was comprehensively investigated through parametric studies, where a Q-factor of 50.72 and a maximum sensitivity of 149.5 GHz/RIU were reported. Meng *et. al* demonstrated a multi-band terahertz absorber made of periodic square metallic ribbon with a T-shaped gap. Several resonator structures like square spiral shape[32], SSRR( square split ring resonator)[33], Jerusalem Cross[34] , CSRR (Circular split ring resonator)[35], eight-resistive arm cell[36], and Ring C-shape quasi-MMA[37] have been investigated. Most of the reported THz meta-sensors suffer from the limitations such as larger unit cell size, one or two absorption peaks, low absorption coefficients, and polarization sensitivity. In this paper, we report a THz metamaterial perfect absorber-based biosensor that shows almost 100% absorption in the THz frequency range for fast and non-destructive detection of microorganisms like molds, yeast cells, fungi, and pesticides. We observe multi-resonant (four) peaks in the frequency range of 0.1 - 3.8 THz. We carried out detailed simulations to study the sensing performance of the proposed sensor. A significant frequency shift in the inductive-capacitive resonance is obtained with the application of microorganisms. A mapping of shifts in the resonance peaks of MPA with structural parameters is carried out. Multi-resonant MPA-based THz biosensors are remarkably desirable for label-free biosensing of a family of microorganisms (bacteria, yeast, molds, etc.). The proposed sensor is extremely suitable for the detection of pesticides because their specific absorption fingerprint lie in the frequency range of 0.1 to 3 THz.

## Design and analysis of MPA based sensor

We investigate several metastructures consisting of 1) only one stripe resonator, 2) two stripes resonator, 3) two stripes and one square resonator (SR), 4) two stripes and two SRs, and 5) two stripes and 1 open book-like resonator (designed from putting two SRs obliquely) on Gallium Arsenide (GaAs) substrate (dielectric constant 12.96) for MPA based sensor (Fig.1). GaAs substrate is selected due to its high band gap, high breakdown voltage, and better heat & moisture resistivity. A thick gold layer on the lower side of the substrate as a ground plane is used to suppress the transmission of the THz signal. We select gold due to its high conductivity, which causes inductance and capacitance to produce resonance at the desired frequency.

The highest absorption coefficient is obtained for an open book-type metastructure, where oblique sheets offer the ability to manipulate the bandwidth of MPA leading to different resonance conditions and different optical paths for resonances[38]. Sensor parameters of an open book-type metastructure are optimized as: periodicity $(P) = 70\mu m$, substrate thickness $(t_s) = 15\mu m$, width of metal stripes $(w_s) = 4.5\mu m$, length of metal stripes $(L_s) = 50\mu m$, gap between the metal stripes $(G) = 40\mu m$, the thickness of ground metallic plane $(t_m) = 0.08\mu m$, length$(L)$ and width$(w)$ of the open book resonator $= 20\mu m$ and $15\mu m$, $g = 5\mu m$ respectively. A plane electromagnetic wave(THz) polarized along the x-axis is used to illuminate the proposed device. We applied unit cell boundary conditions in the x- and y-directions, while the THz beam is incident normally along the z-direction. The schematic of MPA - based sensor with incident, reflected, and absorbed THz beam is shown in figure 1(f). Simulations are carried

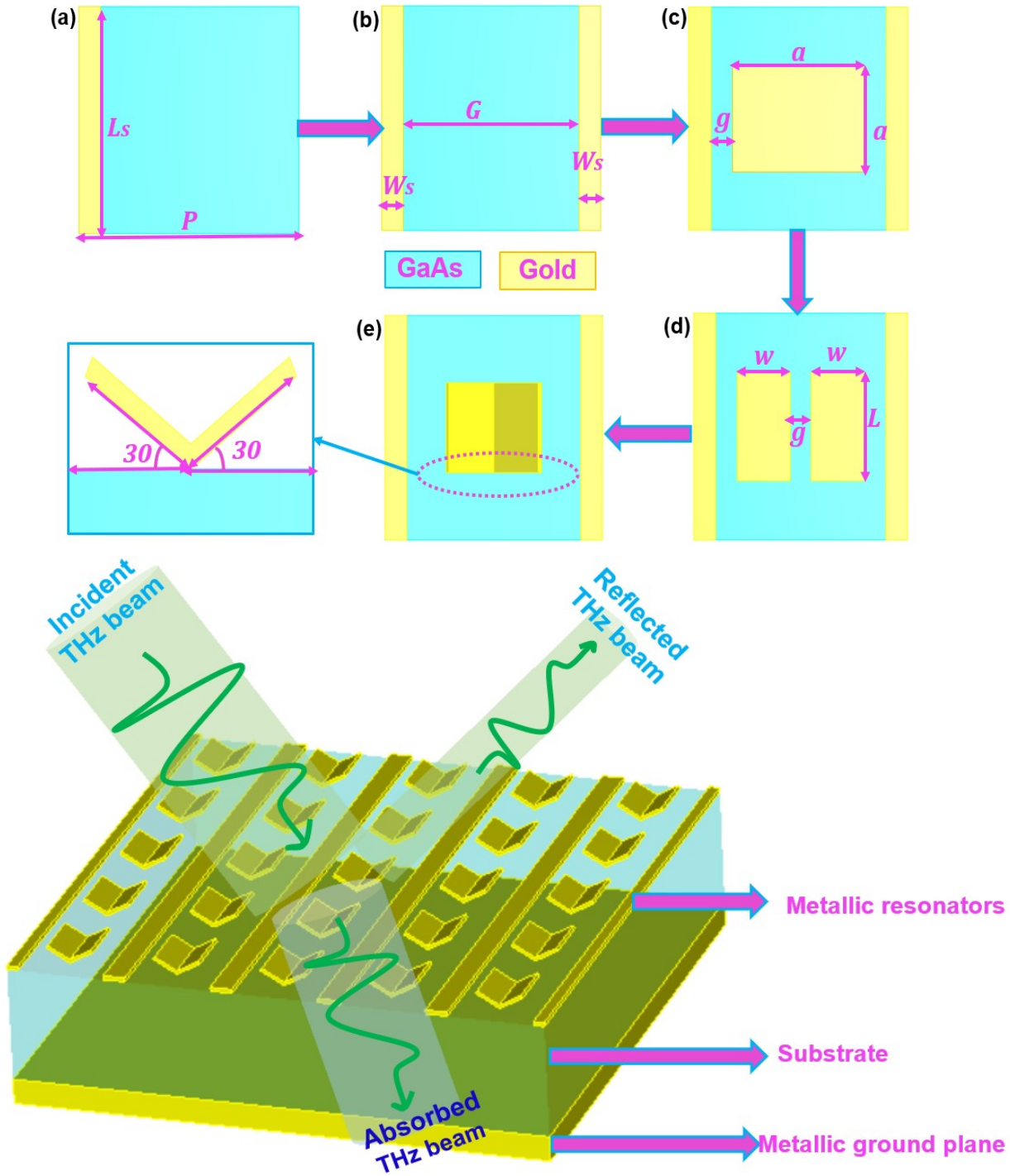



**Figure 1.** Schematic of the unit cell with (a) only one metal stripe resonator, (b) two metal stripes resonator, (c) two stripe +one square resonator(SR), (d) two stripe + two SRs, (e)two stripes + 1 open book-like resonator, which is designed by putting two SRs obliquely., and (f)schematic of MPA - based sensor with incident, reflected, and absorbed THz beam

out using commercial software CST Microwave Studio Suite in the frequency domain. A finite element method-based algorithm is employed to obtain precise simulation results. To assure minimum reflection

of THz beam from metamaterial surface ($R(\omega) = S_{11}^2$), a perfect impedance matching with air intrinsic impedance ($Z_o$) (where real and imaginary parts of $Z(\Omega)$ are close to 1 and 0 respectively) is obtained. To suppress the transmission of THz wave through MPA to zero, a metallic ground plane on the bottom side of substrate ($T(\omega) = S_{21}^2 = 0$) is used. We obtain four absorption peaks in the frequency range 0.1 to $3.7THz$ with almost 100% absorptivity. The absorption frequencies are calculated as

$$A(\omega) = 1 - (R(\omega) + T(\omega)) \tag{1}$$

We calculate the absorption coefficients of MPA as

$$A(\omega) = 1 - R(\omega) = 1 - \left| \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \right|^2 \tag{2}$$

where $Z_{in}$ is the impedance of MPA, which depends on its effective permittivity and permeability. The effective permittivity and permeability are the function of electric and magnetic responses of MPA at the resonance frequencies. In terms of S- parameters, $Z_{in}$ is calculated as

$$Z_{in} = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{(1+S_{11})^2 - S_{21}^2}{(1-S_(11)^2 - S_{21}^2}} \tag{3}$$

the equation for absorption coefficient in terms of S-parameters is

$$A(\omega) = 1 - \left| \frac{\sqrt{\frac{(1+S_{11})^2 - S_{21}^2}{(1-S_{11})^2 - S_{21}^2}} - Z_0}{\sqrt{\frac{(1+S_{11})^2 - S_{21}^2}{(1-S_{11})^2 - S_{21}^2}} + Z_0} \right|^2 \tag{4}$$

by simpyfing equation (4), we get

$$A(\omega) = 1 - \left| \frac{\sqrt{(1+S_{11})^2} - Z_0\sqrt{(1-S_{11})^2}}{\sqrt{(1+S_{11})^2} + Z_0\sqrt{(1-S_{11})^2}} \right|^2 \tag{5}$$

To better understand the absorption behavior, we thoroughly investigate the effect of structural parameters on the resonance frequencies. 2D colormaps in Fig. 2 show the effect of variation of length of the stripe ($L_s$), the width of the stripes ($w_s$), width and position of the open book resonator on the absorption coefficient. The value of $L_s$ is varied from $20\mu m$ to $50\mu m$ as shown in Figure 2(a). Due to the increase in inductance and capacitance of metallic stripes with an increase in $L_s$, a redshift in the resonances are observed with the increase in $L_s$. The maximum absorption is obtained at $L_s = 50\mu m$.

The effect of the width of the metal stripe ($w_s$) (from $4\mu m$ to $10\mu m$) on the absorption coefficient is shown in Figure 2(b). The absorption coefficients for all four resonances become unity at $w_s = 4.5\mu m$, which starts decreasing with an increase in $w_s$, which is due to enhancement in the size of metallic resonators leading to high reflections. We also investigate the effect of the $w$ parameter (width of open book type SRs from $w = 5 - 30\mu m$) on the absorption performances of the MPA(Figure 2(c)). We obtain the highest absorption for $w = 15\mu m$ without any shift in the resonance frequencies. Figure 2(d) shows the effect of the length of SRs (for $L = 10$ to $50\mu m$) on resonance frequencies and absorption coefficients. It is found that the absorption coefficient decreases with the increase in the length of the open book due to high reflections. MPA gives the highest absorption for $L = 20\mu m$ and zero absorption for $L = 50\mu m$.

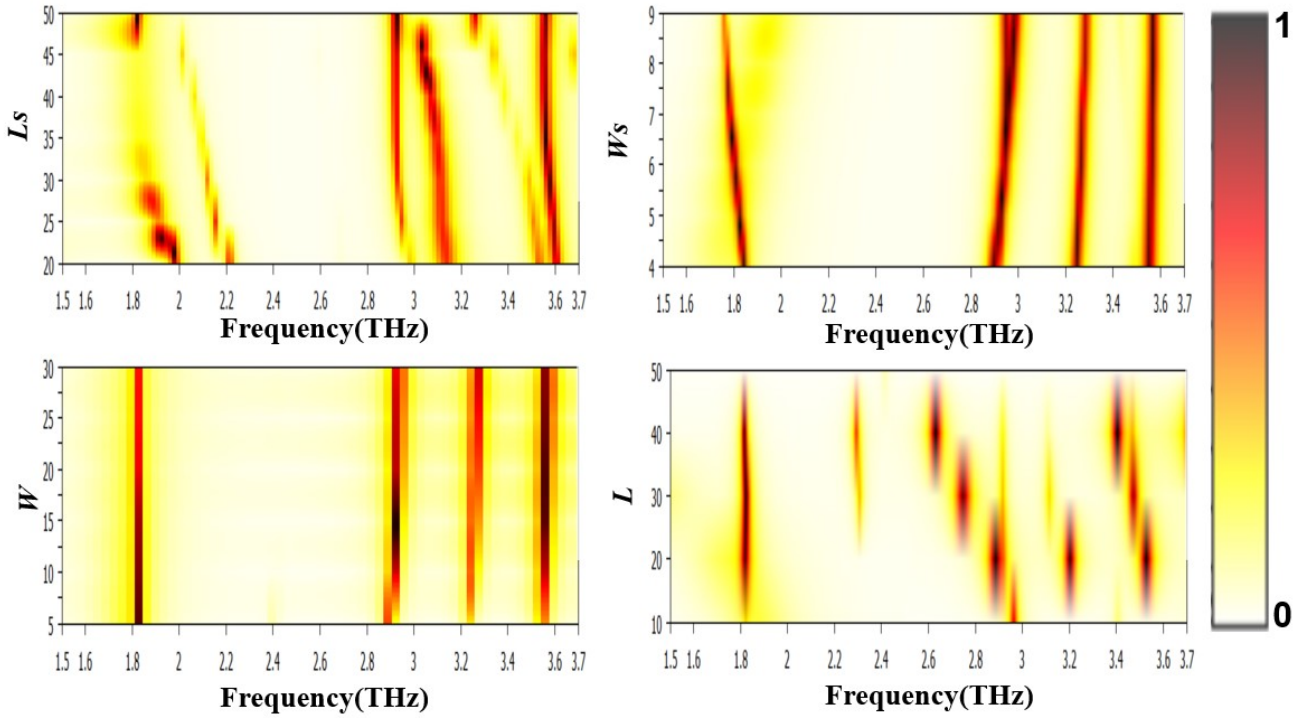



**Figure 2.** 2D colormap absorption characteristics of different (a) length of metal stripe, (b) width of metal stripe, (c) width of SRs of open book resonator, and (d) length of SRs of open book resonator.

## Equivalent circuit model of MPA using Advance Design System (ADS)

To better understand the absorption mechanism of the proposed MPA, an equivalent lumped circuit has been designed using ASD -2019 simulation software (Figure 3), where unit cell of MPA has four resonators i.e. two metal stripes and two SRs as shown in Figure 1(e). Each resonator behaves as an RLC circuit.

The inductance of each metallic resonator $(L_1, L_2, L_3, \& L_4)$ is calculated using Grover's formula[39]

$$L = \mu_0 \mu_r \left( ln\frac{2l}{t_m + w} - 0.2235 ln\frac{t_m + w}{l} + \frac{1}{2} \right) \tag{6}$$

where $l$ is the length of resonators, $w$ is width of resonators, $t_m$ is thickness of resonators. The capacitance of MPA is calculated using[40]

$$c = \varepsilon_o \varepsilon_r \frac{A}{d} \tag{7}$$

where $A, d, \varepsilon_o$ and $\varepsilon_r$ represent the area of the stripe, capacitive gap between the resonators, permittivity of free space, and relative permittivity of the medium, respectively. By tuning the capacitance to get desired resonant frequency, and resistance to increase or decrease the reflection coefficient, we get similar reflection spectra of MPA as obtained in CST microwave studio suite-based simulations. The reflection characteristics $(S_{11})$ obtained by ADS perfectly match with simulation results(figure $3(b)\&(c)$), which confirms the accuracy of the results.

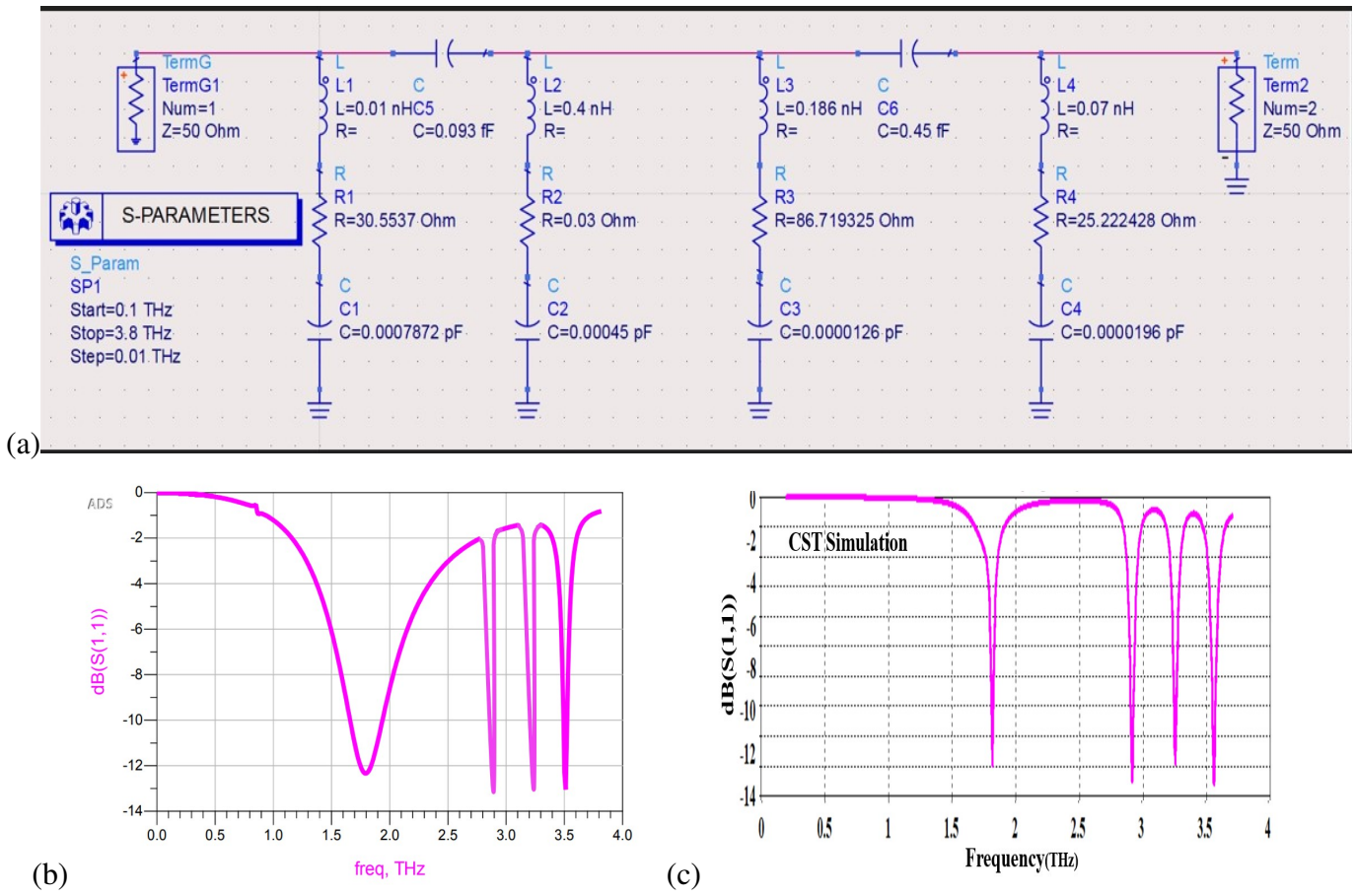



**Figure 3.** (a)Equivalent Circuit design of MPA, (b) reflection coefficient spectra calculated by ADS, (c) reflection coefficient spectra simulated by CST.

## Results and discussion

Prospects of MPA increase with enhancement in the working frequency band and also with the increase in number of absorption peaks. We properly optimize the lattice structure (by optimizing the shape and size of the resonators) of the MPA to enhance working frequency band and also the number of absorption peaks. We carry out simulations for absorption spectra of the proposed MPA for five different structures for : (1) one stripe resonator - we get two resonance frequencies with a maximum of 32% absorption, (2) two stripes resonators- two resonances are obtained but absorption increases up to 50%, (3) two metal stripes and one square resonator (SR)- we get four absorption peaks, first with 100% absorption, second with 70% absorption, third with 58% absorption, and fourth with 55% absorption, (4) two metal stripes and two SRs- we get four absorption peaks, first & second with almost 100% absorption, third with 67% absorption, and fourth with 42% absorption, and (5) two metal stripes and one open book like resonator between the stripes –four resonance absorption peaks at frequencies $f_1 = 1.7THz, f_2 = 2.8THz, f_3 = 3.2THz, f_4 = 3.5THz$ are acheived as shown in figure 4(a). The absorption coefficients obtained forthese resonance frequencies are $99.12\%, 99.95\%, 98.5\%$, and $99\%$, respectively. The high absorption coefficients occur due to the strong inductive–capacitive coupling between the resonators. The resonators of different sizes own the ability to produce absorption peaks at specific frequencies and their superposition effect yields perfect absorption in multiple frequency bands[41–43]. Thus, the dimension of the unit cell structure and the shape of resonators have a vital role in accomplishing absorption near unity.

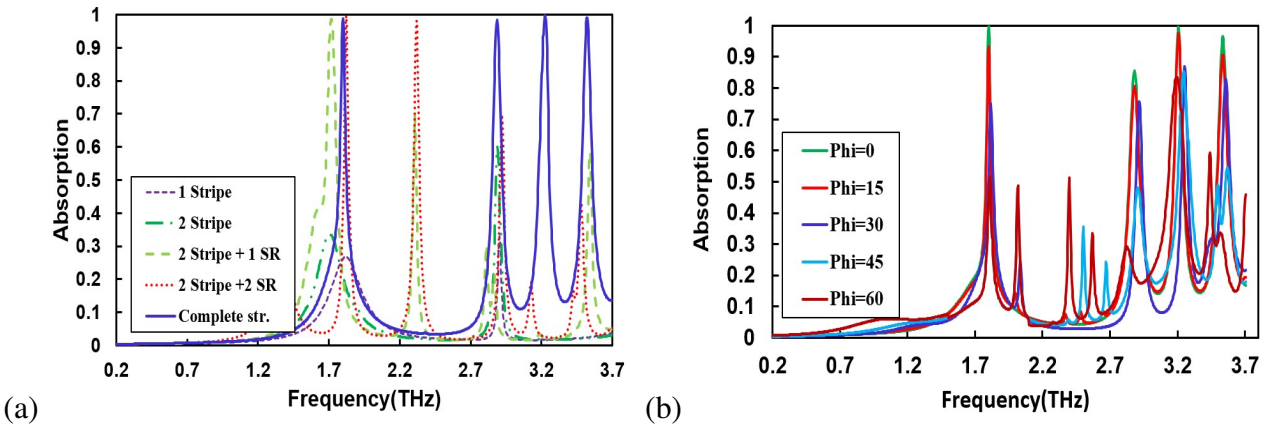



**Figure 4.** (a)The simulated absorbance spectra of the designed MPA for five different types of resonators; (b) Absorption spectra of the MPA with the different oblique incidence of THz beam..

We also figure out the effect of incidence angle on the absorption peaks. Absorption spectra with different angles of incidence of THz wave is shown in Figure 3(b). As the incidence angle increases, the first resonance absorption peak (1.79THz) is almost stable while higher-order absorption peaks display deviations. It shows that higher-order absorption peaks are more sensitive to the incidence angle of THz than the first absorption peak. Thus, the first resonance absorption frequency is a good choice for the target sensing of microorganisms.

We investigate the performance of the sensor for different microorganisms such as yeast, pesticides, and fungi. Experimental data reported by Park *et.al*[14] and theoretical data (obtained using the effective

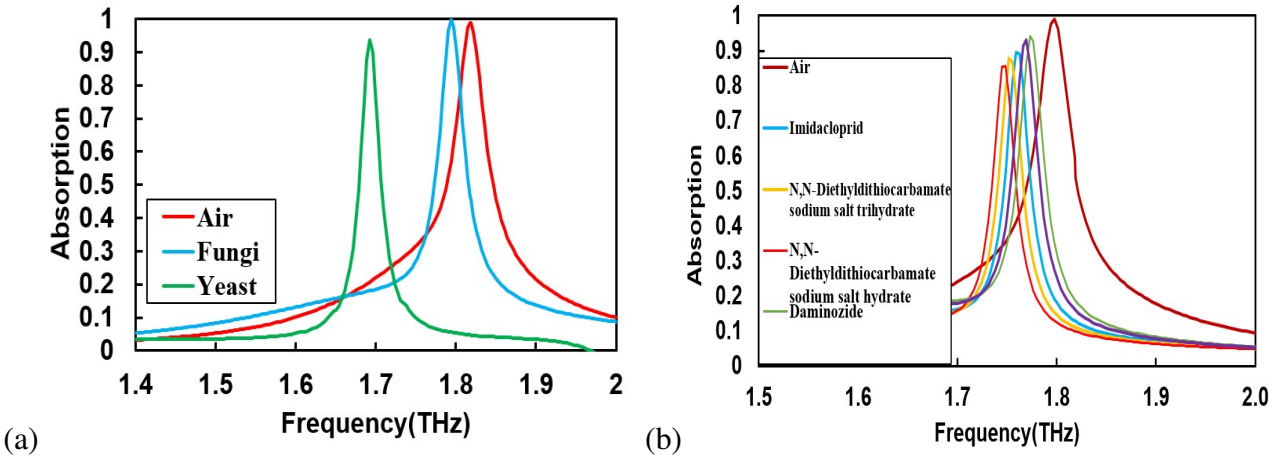



**Figure 5.** (a)Absorbance spectra for Penicillium chrysogenum(blue), yeast(red) and air (green) with effective dielectric constant of 1.94, 8.2 and 1, respectively, and (b)Absorbance spectra for five types of pesticides.

medium theory based on the Maxwell-Garnett mode) for dielectric constants of microorganisms are used for our investigations. Experimentally, Penicillium chrysogenum (fungi) and yeast were grown by streaking on medium method at $37^{\circ}C$ and deposited on the surface of metamaterial. A conventional photo-lithography method was used to prepare the metamaterial pattern and THz time-domain spectroscopic technique was used to characterize the samples[14]. The measured values of the dielectric constants were

reported between 1.37-1.94 for fungi and 5.38-8.54 for yeast. Experimental findings were examined using the finite difference time-domain solver. The absorption patterns for Penicillium chrysogenum(fungai) and yeast are shown in Figure 5(a). The respective frequency shifts of 70GHz and 103GHz are observed for Penicillium chrysogenum and yeast. Due to large dielectric constant of yeast, redistribution of the resonance of electric and magnetic fields leading to a strong modulation of the absorption peak provides larger resonance shift. The traits of the sensed microorganism can be obtained by calculating the frequency shift of the absorption peaks.

We also study the sensing performance for several pesticides like Imidacloprid, N, N-Diethyldithiocarbamate sodium salt trihydrate, Daminozide, N, N-Diethyldithiocarbamate sodium salt hydrate, and Dicofol. For this purpose, we use the experimental values of refractive indices reported by Maeng et. al[5]. The measured refractive index(RI) of Imidacloprid, N, N-Diethyldithiocarbamate sodium salt trihydrate, Daminozide, N, N-Diethyldithiocarbamate sodium salt hydrate, and Dicofol are shown in Table 1. The absorption vs frequency spectra of all pesticides is shown in Figure 5(b). A redshift of resonance frequencies by $70GHz$ is observed in presence of pesticides.

| Name of Pesticides | Formula | RI | Absorption (THz) |
|---|---|---|---|
| Imidacloprid | $C_9H_{10}CIN_5O_2$ | 1.87 | 1.36,1.76 |
| N,N-Diethyldithiocarbamate sodium salt trihydrate | $C_5H_{10}NNaS_2 * 3H_2O$ | 1.99 | 1.11, 1.41, 1.74, 2.40 |
| N,N-Diethyldithiocarbamate sodium salt hydrate | $C_3H_6NNaS_2 * H_2O$ | 2.08 | 0.99, 1.50 |
| Daminozide | $C_6H_{12}N_2O_3$ | 1.68 | 1.32, 2.40 |
| Dicofol | $C_{14}H_9C_{15}O$ | 1.70 | 2.34 |

**Table 1.** Nmae of pesticides used for sensing, along with their molecular formula, refractive index, and absorption peak in THz region

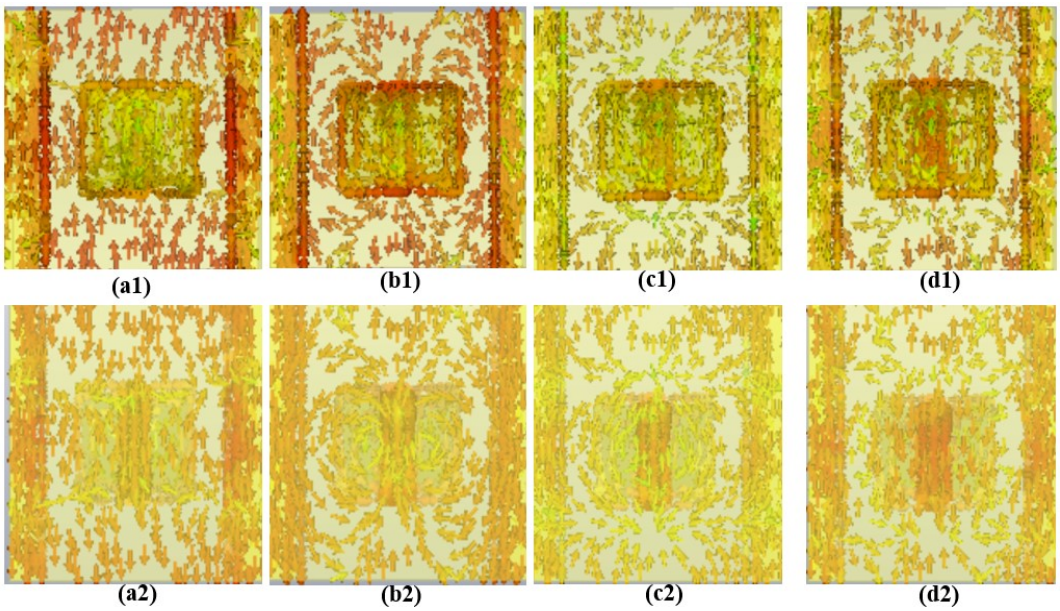



**Figure 6.** Surface current distributions on top layer (a1-d1) and bottom layer (a2-d2) of the designed MPA at resonance frequencies:
$(a1,a2)f_1 = 1.7THz, (b1,b2)f_2 = 2.8THz, (c1,c2)f_3 = 3.2THz, (d1,d2) and f_4 = 3.5THz.$

To gain deep insight into the physical mechanism behind the absorption of MPA, we studied the surface current distribution on the top and bottom metallic layers of the unit cell at resonance conditions. Figure. 6(a1-d1) & (a2-d2) show the surface current distribution of the top and the bottom layers, respectively. Antiparallel currents form a current loop leading to the excitation of the magnetic dipole resonance[44]. The surface currents on both metallic layers are flowing in opposite directions at resonance (Fig. 6), which generates the magnetic flux coupling with the incident H-field. The antiparallel direction of surface currents on the top and bottom layers of MPA validates that the absorption peaks are caused by magnetic resonance responses. Therefore, the stronger absorption of the MPA is instigated by the excitation of multiple high-order magnetic resonances.

## Conclusion

We develop a novel THz MPA-based sensor for sensing of biological analytes. The highest absorption coefficient is obtained for an open book-type metastructure. Four resonance absorption peaks at frequencies $f_1 = 1.7THz, f_2 = 2.8THz, f_3 = 3.2THz, f_4 = 3.5THz$ having absorption coefficients $99.12\%, 99.95\%, 98.5\%$, and $99\%$, respectively are obtained.We also study the absorption spectra of the MPA with different oblique incidences of THz wave and found that lower resonance absorption peak $(1.79THz)$ is almost stable while higher absorption peaks show deviations. The effect of structural parameters(length and width of metal stripes and book resonator) on the resonance frequencies is analysed to get the optimum performance of the MPA. To gain deep insight into the mechanism of MPA, an equivalent circuit model is developed using Advance Design System (ADS) software. Surface current distribution depicts that strong absorption of the MPA is instigated by the excitation of multiple high - order resonances.Due to the change in the effective dielectric constant of MPA environment with the deposition of microorganisms, significant redshifts in the resonance peaks (redshift of $103GHz$ for yeast, redshift of $70GHz$ for fungi) are observed. These shifts in resonance frequencies are due to the redistribution of the resonance electric and magnetic fields, which lead to a strong modulation of the absorption. Thus, by functionalizing the substrates with antibodies a selective detection of microorganisms is also plausible. We believe that this MPA-based THz biosensor can open up new opportunities for obtaining dielectric information of microorganisms, which has great potential in label-free biosensing for future biomedical applications.

## References


1. Kagan, C. R. At the nexus of food security and safety: opportunities for nanoscience and nanotechnology (2016).
2. Feng, Y.-Z. & Sun, D.-W. Application of hyperspectral imaging in food safety inspection and control: a review. *Critical reviews food science nutrition* **52**, 1039–1058 (2012).
3. Bhati, R., Jewariya, M. & Malik, A. K. Independently tunable quint band electromagnetically induced transparency windows for multifunctional terahertz device. *Appl. Phys. A* **129**, 436 (2023).
4. Qin, J., Xie, L. & Ying, Y. Determination of tetracycline hydrochloride by terahertz spectroscopy with plsr model. *Food chemistry* **170**, 415–422 (2015).
5. Maeng, I. *et al.* Feasibility of using terahertz spectroscopy to detect seven different pesticides in wheat flour. *J. Food Prot.* **77**, 2081–2087 (2014).
6. Ferguson, B. & Zhang, X.-C. Materials for terahertz science and technology. *Nat. materials* **1**, 26–33 (2002).

**7.** George, P. A. *et al.* Microfluidic devices for terahertz spectroscopy of biomolecules. *Opt. Express* **16**, 1577–1582 (2008).

**8.** Park, H.-R. *et al.* Colossal absorption of molecules inside single terahertz nanoantennas. *Nano letters* **13**, 1782–1786 (2013).

**9.** Mickan, S. P. *et al.* Label-free bioaffinity detection using terahertz technology. *Phys. Medicine & Biol.* **47**, 3789 (2002).

**10.** Menikh, A., Mickan, S. P., Liu, H., MacColl, R. & Zhang, X.-C. Label-free amplified bioaffinity detection using terahertz wave technology. *Biosens. Bioelectron.* **20**, 658–662 (2004).

**11.** Bhati, R., Jewariya, M. & Malik, A. K. Spoof surface plasmon-based terahertz metasensor for glucose and ethanol. *Appl. Phys. A* **128**, 840 (2022).

**12.** Berrier, A., Schaafsma, M. C., Nonglaton, G., Bergquist, J. & Rivas, J. G. Selective detection of bacterial layers with terahertz plasmonic antennas. *Biomed. optics express* **3**, 2937–2949 (2012).

**13.** Park, S., Son, B., Choi, S., Kim, H. & Ahn, Y. Sensitive detection of yeast using terahertz slot antennas. *Opt. Express* **22**, 30467–30472 (2014).

**14.** Park, S. *et al.* Detection of microorganisms using terahertz metamaterials. *Sci. reports* **4**, 1–7 (2014).

**15.** O'Hara, J. F. *et al.* Thin-film sensing with planar terahertz metamaterials: sensitivity and limitations. *Opt. Express* **16**, 1786–1795 (2008).

**16.** Bhati, R. & Malik, A. K. Ultra-efficient terahertz metamaterial sensor. *Results Opt.* **8**, 100236 (2022).

**17.** Ghobadi, A., Hajian, H., Gokbayrak, M., Butun, B. & Ozbay, E. Bismuth-based metamaterials: from narrowband reflective color filter to extremely broadband near perfect absorber. *Nanophotonics* **8**, 823–832 (2019).

**18.** Keshavarz, A. & Zakery, A. A novel terahertz semiconductor metamaterial for slow light device and dual-band modulator applications. *Plasmonics* **13**, 459–466 (2018).

**19.** Shen, S. *et al.* Recent advances in the development of materials for terahertz metamaterial sensing. *Adv. Opt. Mater.* **10**, 2101008 (2022).

**20.** Cui, Y. *et al.* Plasmonic and metamaterial structures as electromagnetic absorbers. *Laser & Photonics Rev.* **8**, 495–520 (2014).

**21.** Watts, C. M., Liu, X. & Padilla, W. J. Metamaterial electromagnetic wave absorbers. *Adv. materials* **24**, OP98–OP120 (2012).

**22.** Hasan, D. & Lee, C. Hybrid metamaterial absorber platform for sensing of co2 gas at mid-ir. *Adv. Sci.* **5**, 1700581 (2018).

**23.** Ren, Z., Lin, Z., Zhi, X. & Li, M. Double-layer broadband perfect metamaterial absorber and its potential for refractive index sensing. *Opt. Mater.* **99**, 109575 (2020).

**24.** Saadeldin, A. S., Hameed, M. F. O., Elkaramany, E. M. & Obayya, S. S. Highly sensitive terahertz metamaterial sensor. *IEEE Sensors J.* **19**, 7993–7999 (2019).

**25.** Tao, H. *et al.* Performance enhancement of terahertz metamaterials on ultrathin substrates for sensing applications. *Appl. Phys. Lett.* **97**, 261909 (2010).

**26.** Areed, N. F., El-Wasif, Z. & Obayya, S. Nearly perfect metamaterial plasmonic absorbers for solar energy applications. *Opt. Quantum Electron.* **50**, 1–12 (2018).

**27.** Tan, S., Yan, F., Wang, W., Zhou, H. & Hou, Y. Ultrasensitive sensing with three-dimensional terahertz metamaterial absorber. *J. Opt.* **20**, 055101 (2018).

**28.** Cong, L. *et al.* Experimental demonstration of ultrasensitive sensing with terahertz metamaterial absorbers: A comparison with the metasurfaces. *Appl. Phys. Lett.* **106**, 031107 (2015).

**29.** Zamzam, P. & Rezaei, P. A terahertz dual-band metamaterial perfect absorber based on metal-dielectric-metal multi-layer columns. *Opt. Quantum Electron.* **53**, 1–9 (2021).

**30.** Wang, J., Lang, T., Hong, Z., Xiao, M. & Yu, J. Design and fabrication of a triple-band terahertz metamaterial absorber. *Nanomaterials* **11**, 1110 (2021).

**31.** Abdulkarim, Y. I. *et al.* Simulation and lithographic fabrication of a triple band terahertz metamaterial absorber coated on flexible polyethylene terephthalate substrate. *Opt. Mater. Express* **12**, 338–359 (2022).

**32.** de Araújo, J. B. O. *et al.* An ultrathin and ultrawideband metamaterial absorber and an equivalent-circuit parameter retrieval method. *IEEE Transactions on Antennas Propag.* **68**, 3739–3746 (2020).

**33.** Hannan, S. *et al.* Modified-segmented split-ring based polarization and angle-insensitive multi-band metamaterial absorber for x, ku and k band applications. *IEEE Access* **8**, 144051–144063 (2020).

**34.** Jain, P. *et al.* Ultra-thin metamaterial perfect absorbers for single-/dual-/multi-band microwave applications. *IET Microwaves, Antennas & Propag.* **14**, 390–396 (2020).

**35.** Hannan, S., Islam, M. T., Almutairi, A. F. & Faruque, M. R. I. Wide bandwidth angle-and polarization-insensitive symmetric metamaterial absorber for x and ku band applications. *Sci. Reports* **10**, 1–9 (2020).

**36.** Nguyen, T. T. & Lim, S. Design of metamaterial absorber using eight-resistive-arm cell for simultaneous broadband and wide-incidence-angle absorption. *Sci. reports* **8**, 6633 (2018).

**37.** Hoque, A. *et al.* A polarization independent quasi-tem metamaterial absorber for x and ku band sensing applications. *sensors* **18**, 4209 (2018).

**38.** Lu, C.-Y., Chung, C.-C., Yen, T.-J. & Huang, T.-Y. Achieving broad absorption band and high incident angles by stochastically-distributed oblique-flat-sheet metamaterial perfect absorbers. *Sci. Reports* **11**, 18713 (2021).

**39.** Grover, F. W. *Inductance calculations: working formulas and tables* (Courier Corporation, 2004).

**40.** Hakim, M. L. *et al.* Polarization insensitive symmetrical structured double negative (dng) metamaterial absorber for ku-band sensing applications. *Sci. reports* **12**, 1–18 (2022).

**41.** Zhang, B., Hendrickson, J. & Guo, J. Multispectral near-perfect metamaterial absorbers using spatially multiplexed plasmon resonance metal square structures. *JOSA B* **30**, 656–662 (2013).

**42.** Zhang, J., Tian, J. & Li, L. A dual-band tunable metamaterial near-unity absorber composed of periodic cross and disk graphene arrays. *IEEE Photonics J.* **10**, 1–12 (2018).

**43.** Wang, B.-X., Tang, C., Niu, Q., He, Y. & Chen, T. Design of narrow discrete distances of dual-/triple-band terahertz metamaterial absorbers. *Nanoscale research letters* **14**, 1–7 (2019).

**44.** Liu, N. & Giessen, H. Coupling effects in optical metamaterials. *Angewandte Chemie Int. Ed.* **49**, 9838–9852 (2010).

## Acknowledgements

The author Ruchi Bhati thanks CSIR, Government of India for her research fellowship.

## Author contributions statement

R.B developed the concept, carried out the simulation, theoretical model, analysis of data, and wrote the manuscript. A.K.M., as the principal investigator, conceived the idea, planned, coordinated and supervised the project.

## Competing interests

The authors declare no competing financial interests.

## Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

## Methods

We have used commercial software CST Microwave studio suite to design and simulate our MM unit cell. In the x – and y – directions, unit cell boundary conditions are applied and in z-direction, open add space is employed to obtain the absorption behavior of the unit cell with adaptive tetrahedral messing. THz radiation is incident normally on the structure with an electric field along the x- direction. A field monitor is applied at the resonant frequency to obtain electric field and surface current distribution. Additionally, advanced design system(ADS -2019) software is used to design equivalent circuit of MPA using to analysis its functioning. In our simulation, we chose GaAs as a substrate with a dielectric constant of 12.94 and loss tangent of 0.006, and gold as a metal for resonator[16].

## Supplementary Files

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